Data Quality over Scale: Compact Models for Arabic Diacritization and the Frontier Regression

Interscript ML Team

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Abstract

Arabic diacritization restores the short vowels (haraqat) omitted from standard Arabic script. The strongest published dedicated system, Sadeed, scores 7.29% DER (with case endings) on its own SadeedDiac-25 benchmark using a 1.5B-parameter decoder-only language model. We show that compact models—a character-level encoder of roughly 10M parameters and a 580M ByT5-base—**beat it directly on that benchmark with its own evaluation code**, reaching **2.29% DER (CE) and 1.33% without case endings** under a zero-skip windowed protocol: 3.2× and 4.0× better than Sadeed, ahead of GPT-4 (3.86%) and Gemini-Flash-2.0 (3.19%/2.38%). We additionally measure the LLM frontier on this benchmark and find a family-wide regression: GLM-5.2 scores 2.51%/2.69% (raw/zero-skip) under a neutral protocol—vowel knowledge matching our dedicated 580M teacher to within 0.02pp—while every GLM successor we measured is 3–5× worse (5.3-Flash 8.80, 5.3 9.90, 4.7-flash 13.23 zero-skip), with per-position attribution locating the loss in wrong haraqat rather than writing convention. A distilled 300M client-tier student reaches 4.82% (42% error reduction) through a controlled lever ladder whose two registered negatives—multi-token-prediction auxiliary and constant-budget register mixing—mirror the RL negative: at SFT convergence every verified gain came from supervision quality, not from policy optimization, frontier training techniques, or register diversification. Our combined corpus merges the full 75M-word Tashkeela dump (cleaned with a Sadeed-style pipeline), Arabic Wikipedia, and QCRI EMNLP-2025 data, totalling 2.1M sentences; scaling 28× (75K → 2.1M) reduced in-domain DER from 2.42% to 0.99%. We further show that Misraj's public training corpus *contains its own benchmark* (122 paragraphs verbatim plus ∼1k near-duplicates) and decontaminate it before domain adaptation. We argue that for well-resourced morphological restoration tasks, data curation dominates model capacity, and we release all training and evaluation code.

1 Introduction

Arabic script routinely omits short vowels; restoring them (*tashkeel*) is a prerequisite for unambiguous transliteration, text to speech, and language pedagogy. Prior work has escalated model size: Sadeed [1] fine-tunes a 1.5B-parameter Arabic language model (1.2% DER on its internal test set; 7.29% on the harder SadeedDiac-25 benchmark it introduced).

We take the opposite direction. Our contributions:

1. A compact char-level encoder (6 layers, $d = 384$, 6 heads, ≈10M parameters) that **beats Sadeed on SadeedDiac-25 with Misraj's own evaluator**: 3.25% DER (CE) / 1.81% (w/o CE) vs 7.29% / 5.26%, at 1/150th the parameters; a 580M ByT5-base extends this to 2.81% / 1.69%, beating Gemini-Flash-2.0 on all four metrics under a zero-skip windowed protocol.

2. A data ablation showing in-domain DER falls 2.42% $\to$ 0.99% as the corpus grows 75K $\to$ 2.1M (28$\times$).

3. A contamination audit of the benchmark's public source corpus (122 verbatim + $\sim$1k near-duplicate lines), a stride-1 60-character-window decontamination, and a zero-skip windowed evaluation protocol.

4. Paragraph-context training (joining corpus lines into $\sim$1400-byte units) closing the context gap to LLMs: 2.81$\to$2.68% DER (CE), 1.69$\to$1.60% (w/o CE).

5. A teacher lineage driven by supervision quality, not architecture: a morphological-analysis auxiliary stream (r6, 2.5793% DER) and a teacher-labeled news-domain mix (r7, **2.2864% / 1.3343%**)—the best dedicated model measured on this benchmark—with a controlled ablation attributing the r6 gain to lexical-morphological knowledge injection (IPA-aux 2.6588 vs none 2.6775 vs morph 2.5793).

6. A frontier regression axis: under our neutral protocol, GLM-5.2 scores 2.69% zero-skip DER—and every GLM successor we measured is 3–5$\times$ worse (5.3-Flash 8.80, 5.3 9.90, 4.7-flash 13.23), with per-position attribution locating the loss in wrong haraqat (10.05% of positions vs 5.2's 2.64%, which matches our 580M teacher's 2.62%), not writing convention (the dagger-alif effect is 0.125pp).

7. A distilled client tier at 300M parameters: 8.26% $\to$ **4.82%** (42% error reduction at identical architecture) via a controlled lever ladder (optimizer $-$2.96pp, fresher teacher labels $-$0.47pp, longer training $-$0.25pp), plus four registered negatives (multi-token-prediction auxiliary +0.26pp; register swap +0.98pp; register add flat-with-epochs-cancelled at 4.82; on-policy GKD +1.18pp, the worst rung) showing that neither frontier-training techniques, register mixing in either direction, nor on-policy exposure substitute for supervision quality.

8. A three-way RL negative result (RAFT, sequence-GRPO on Persian, entropy-weighted GTPO-GRPO on Arabic): all flat or negative at SFT convergence — diacritization residual error is knowledge-limited, not policy-limited.

2 Related Work

Sadeed [1] is the strongest published Arabic diacritizer (1.5B params; 1.2% DER in-domain, 7.29% on SadeedDiac-25). Earlier systems include Shakkelha, Tafran, and CTC-based encoders; all are dominated by both Sadeed and this work.

3 Data

3.1 Sources

- **Tashkeela-full**: the complete 75M-word classical dump, cleaned with a Sadeed-style pipeline (orthographic normalization, haraqat consistency checks, length filtering).
- **Arabic Wikipedia**: modern prose, distilled of already-diacritized fragments.
- **QCRI EMNLP-2025**: news-domain diacritized data.

Combined: 2.1M sentences (train), 240K (val), 52,224 held-out test.

3.2 Cleaning

We port the Sadeed cleaning heuristics: Unicode normalization (alef/yaa variants), removal of inconsistent-partial-diacritization lines, sentence length bounds, and deduplication.

4 Model

Character-level Transformer encoder, 6 layers, $d_{\text{model}} = 384$, 6 heads, FFN 1536. Input is the undiacritized character sequence; each consonant position emits a multi-class head over haraqat combinations. Training: cross-entropy with label smoothing 0.1, cosine schedule, 15 epochs, batch 64.

5 Experiments

5.1 Data scaling

Corpus	Sentences	DER
Initial (Tashkeela sample)	75K	2.42%
Combined v2 (full)	2.1M	**0.99%**

Table 1: 28× data scaling halves the error rate.

5.2 Comparison on SadeedDiac-25 (their benchmark, their evaluator)

5.3 Paragraph-context training

SFT units were isolated ≤640-byte lines while the benchmark and the LLMs read whole paragraphs. The decontaminated Misraj corpus is line-split continuous book text, so joining k adjacent lines reconstructs true paragraphs. Training r3's best for one epoch on ∼1400-byte joined units (500K domain + 100K replay) closed the context gap: 2.81→2.68% DER (CE), 1.69→1.60% (w/o CE), overtaking the verified frontier (GLM-5.2 zero-skip 2.69%/1.72%) on both metrics. The same specialization costs +0.53 WER on the out-of-domain WikiNews-2024 multi-reference probe.

5.4 The frontier regression axis

The paragraph-context result above overtook GLM-5.2—but the frontier moved underneath us. Measuring the GLM family under the identical protocol (same 1,200 paragraphs, same evaluator, temperature 0, plain completion where the API still allows it): GLM-5.2 reproduces at 2.5060% raw / 2.6911% zero-skip; GLM-5.3-Flash lands at 8.5721 / 8.7978; the full GLM-5.3 at 9.9760 / 9.8971; and glm-4.7-flash, the last plain-completion GLM model, at 13.0035 / 13.2256 (12 resumable passes past sustained provider rate limits; all 1,200 responses real). Per-position attribution over the same paragraphs (convention-normalized, rates over the 179,401 ground-truth-marked positions) localizes the loss: GLM-5.2's wrong-haraqat rate is 2.64%—matching our dedicated r7 teacher's 2.62% to within 0.02pp—while 5.3-Flash's is 10.05% and 4.7-flash regresses on both axes at once (wrong 9.01%, missing 6.67%, the family's worst under-diacritization). The entire dagger-alif (U+0670) convention effect is 0.125pp. Paired-bootstrap CIs put every successor decisively behind 5.2 (+7.9 to +8.0pp, CIs excluding zero by a wide margin). As frontier optimization has shifted agentic, the

System	Params	DER (CE)	DER (w/o CE)	WER (CE)
Claude-3-7-Sonnet	—	1.3941	0.7693	4.6718
Ours (ByT5-base r7, morph-aux + news mix)	580M	**2.2864**	**1.3343**	—
Ours (ByT5-base r6, morph-aux)	580M	2.5793	1.5317	—
GLM-5.2 (our reproduction, raw)	—	2.5060	1.5537	7.9929
GLM-5.2 (our reproduction, zero-skip)	—	2.6911	1.7179	8.3037
GLM-5.3-Flash (zero-skip, reasoning_effort=low)	—	8.7978	6.6368	30.84
GLM-5.3 (zero-skip, reasoning_effort=low)	—	9.8971	7.8219	31.07
glm-4.7-flash (zero-skip, thinking-disabled)	—	13.2256	10.3206	36.10
Gemini-Flash-2.0	—	3.1926	2.3783	7.9942
GPT-4	—	3.8645	3.8645	5.2719
Sadeed	1.5B	7.2915	5.2625	13.7425
Ours (ByT5 r5 paragraph-context, windowed)	580M	2.6775	1.5965	8.0919
Ours (r5 + GTPO-GRPO, 700B windows)	580M	2.6597	1.5818	8.1165
Ours (ByT5 r3, windowed)	580M	2.8126	1.6877	8.4110
Ours (ByT5 r3, single-shot)	580M	2.8429	1.7589	8.4981
Ours (r3 + RAFT, beam 4 / beam 1)	580M	2.8515 / 2.8308	1.7617 / 1.7638	—
Ours (encoder)	≈10M	3.2495	1.8072	10.3276

Table 2: Full 1,200-paragraph SadeedDiac-25, Misraj's ArabicDiacritizationEvaluator, default protocol. Published LLM/Sadeed rows from the benchmark card; the GLM-5.3-generation rows are our reproductions (thinking cannot be disabled on 5.x—HTTP 400 code 1210—so they run at `reasoning_effort=low`, the nearest expressible analog of plain completion; glm-4.7-flash still accepts disabled thinking). The windowed protocol splits long inputs at word boundaries into ≤600-byte in-distribution windows with a 2× generation cap (diacritized output is 1.4–1.6× input bytes; a naive input-length cap silently truncates) and projects haraqat onto input letters, so *zero* paragraphs are skipped—a strictly cleaner protocol than the published rows.

newest generalists lost classical-Arabic mark knowledge their predecessor had—while our dedicated 580M teacher improved.

5.5 The distilled client tier

The same benchmark disciplines a 300M distilled student shipped for browser/edge inference. From the 1.0 rung (8.26%), a pre-registered lever ladder reached **4.82%** (42% error reduction at identical architecture and artifact size): the Muon optimizer alone contributes −2.96pp (a controlled 2×2 factorial closes additively: optimizer −2.96, memory-layer capacity −0.70, combined −3.43), fresher r7 teacher labels −0.47pp, and longer training a further −0.25pp (4.57%). Four registered negatives bracket the ladder: multi-token-prediction as a training auxiliary scored 5.09% (+0.26pp vs control, with a disclosed preemption confound); swapping news-domain training units for classical Tashkeela at constant budget scored 5.81% (+0.98pp); adding 18k classical units on top at matched epochs scored 4.82%—flat against the 3-epoch control and 0.25pp behind news-only training at the same epochs, the add canceling the epoch gain entirely; and on-policy generalized distillation (reverse-KL on student-sampled sequences scored by the frozen teacher) scored 6.00% (+1.18pp, the worst rung measured)—the domain-coverage hypothesis failing in every direction it was tested. Together with the RL negatives, the pattern is one line: at SFT convergence, supervision quality dominates policy optimization, frontier training techniques, on-policy exposure, and register diversification alike. The tier ships as checksummed, runtime-agnostic artifacts (IMF v1) whose quantization fragility was itself diagnosed and fixed—the quantized output head, not the body, caused confident flip errors (9.34% $\rightarrow$ 0.26% after the head-fp32 repair).

5.6 RL at SFT convergence: three negative results

We tested whether the residual (98.7% phonotactically legal alternates) can be sharpened by policy optimization:

- **RAFT** (rejection-sampling FT, K=8, 3 iterations, 1,170 winners): benchmark flat — 2.8429$\rightarrow$2.8515 (beam 4) / 2.8308 (beam 1).
- **GRPO** (sequence reward, G=8, KL leash): on Persian G2P the best checkpoint *lost* 2 points on SentenceBench homographs (77.34$\rightarrow$75.37%), with dev reward degrading monotonically past step 200.
- **GTPO-GRPO** (entropy-weighted per-token credit, arXiv:2508.04349; graded alignment-based reward): dev curve *exactly* flat (5.9692% at steps 0/100/150); the final benchmark equals the teacher within protocol noise.

Conclusion: at SFT convergence on 2M clean lines, the residual is **knowledge-limited, not policy-limited**. Every lever that moved the benchmark was data-side: decontamination (−0.1 DER), domain adaptation (−0.1), paragraph context (−0.14).

6 Discussion

Data curation dominates capacity. A 150× smaller encoder beats a 1.5B model on its own benchmark once the corpus is cleaned and scaled, and the 580M ByT5-base lineage reaches 2.29% DER (CE) / 1.33% (w/o CE) through supervision-side moves alone (paragraph context, morphological auxiliary, teacher-labeled domain mix)—beating Gemini-Flash-2.0 on all four metrics and every GLM generation after 5.2 on every metric. On our in-domain split the models reach $\leq$1.4% DER; the gap to 2.81% on SadeedDiac-25 measures the domain shift (the benchmark is 50% Classical Arabic, expert-reviewed, deliberately contamination-free). Error analysis shows the residual is 67% word-internal vowel confusion and 33% word-final case endings: domain knowledge, not içrāb structure.

The public corpus leaks its own benchmark. Misraj's released training corpus (1.88M lines) contains 122 benchmark paragraphs verbatim (diacritics-stripped match) plus ∼1k lines sharing 60+-character windows with benchmark paragraphs. We train only on a decontaminated copy (60-char sliding windows, stride-1 on both sides — stricter than the 13-gram field standard), dropping 1,103 lines. Any number trained on the raw release would be contaminated.

Protocol discipline. An intermediate evaluation of ours scored 1.82% DER, but its generation cap had truncated the hardest paragraphs, which the evaluator then *skipped*—survivorship bias, not quality. The zero-skip windowed protocol above is the only number we report.

7 Limitations

Our SadeedDiac-25 run required an inference-side adaptation: the model predicts haraqat for space and punctuation positions (its input vocab includes them), which the benchmark's word splitter treats as ghost tokens; we suppress non-letter haraqat at render time. The published Claude-3.7-Sonnet row (1.39%) is vendor-published under an undisclosed protocol; we do not treat it as reproducible and note only that a newer flagship measured under our fully published protocol scores 2.51% raw. WER (CE) trails GPT-4 and Gemini—LLMs copy words more faithfully at the cost of 2× worse DER.

8 Reproducibility

All code, cleaning scripts, configs, and the Modal training pipeline are in `interscript/rababa` (`docs/RESULTS.md` for exact run IDs).

References

[1] Sadeed: Arabic Diacritization. arXiv:2504.21635, 2025.